

CommitGuru - Improved

Ingests and analyzes code repositories.

=> This tool is improved based on tool Commit Guru, which improved the feature extraction part.

Installation

1. Clone this repository in to an empty directory
2. Check and modify `./config.json`, especially the database config.

Db: information relating to your postgresql database setup

logging: information about how to write logging information

gmail: gmail account to be used to send cas notifications

repoUpdates: how often repositories should be updated for new commits

system: how many worker threads the cas system can use to analyze and ingest repos.

Dependencies

make share you are running in **Linux**

Additional Instructions are available in SETUP.md

- Python ≥ 3.3 and ≤ 3.6
- Pip for Python Version > 3.3 and < 3.6
- Git > 1.7
- R
- python-dev
- rpy2
- requests
- dateutil
- sqlalchemy
- py-postgresql
- GNU grep
- MonthDelta

Installing rpy2

```
pip install rpy2
```

Additional Pip Packages

Install the following packages by doing `pip install` and then the package name. Make sure you are using python3, such as using a virtualenv if using Ubuntu.

- SQL Alchemy (sqlalchemy)
- psycpg2

- requests (requests)
- python-dateutil (python-dateutil)
- any package the runtime pip prompts you that are missing.

To install the MonthDelta package, simply do: `pip install`

<https://pypi.python.org/packages/source/M/MonthDelta/MonthDelta-1.0b.tar.bz2>

Attetion

make sure that this project need the dataset tool PostgreSQL

First-Time Database Setup

Set up the database for the first time by running `python script.py initDb`

Usage

In a terminal, type ``nohup python script.py &`` to start the code repo analyzer and run it in the background.

OFFLine

Update by Zuowei Chen 2024-10-28

I have found if we once use the script.py to extract datasets, everytime it will use **git clone** to clone the project. However, the clone process may be interrupted due to the network problem and we need restart this tool again, which is very hassle and waste of time, so I have integrates a offline version of this tool, the usage can be seen below:

Base

```
python script.py initDb
python create_repositories.py {project_name} {url} #In offline version, {url} can
be any string since we do not use it.
python offline.py {project_name}#make sure that we have already clone the project
to ./CASRepos/git/
```

Mode

```
python offline.py {project_name} {Mode} #Mode can be chosen from "New" and "Old".
"New" is our method and "Old" is the original Commit Guru (Chronological order).
You can not input param 'Mode' and it will run default mode "Old".
```

GUI

This feature was added in version 2.0. This is an extension of the offline mode.

```
python runGUI.py
```

GUI display

The screenshot shows a window titled "Expert Feature Extraction" with standard window controls. Inside, the title "Expert Feature Extraction" is centered, followed by the instruction "Please input the params and click start button." Below this is the "Input Params" section, which contains a "Project Name:" text box (annotated with "input target project") and a "Mode:" dropdown menu (annotated with "choose extraction type") showing "New" and "Old" options. A "start" button is positioned below the mode dropdown. The "Stats:" section shows "Waiting for input" (annotated with "showing the stats now"). The "Process result." section contains an "Output path:" text box (annotated with "output path"). At the bottom is a "log" section (annotated with "exception log") with a scrollable text area.

- Project Name: The target project, make sure you have cloned it to `CASRepos/git/`.
- Mode: `NEW` means BDFE and `Old` means simple chronological order.
- start: If you have prepared the params above, click it and start.
- Output path: The data will be extracted to this path.

Success

Expert Feature Extraction

– □ ×

Expert Feature Extraction

Please input the params and click start button.

Input Params

Project Name:

lab1

Mode:

New

▼

start

Stats:

Process success!

Process result.

Output path:

/home/czw/datasets/lab1.csv

log

\Output path: /home/czw/datasets/lab1.csv

⋮

Error

Expert Feature Extraction

– □ ×

Expert Feature Extraction

Please input the params and click start button.

Input Params

Project Name:

lab12

Mode:

New

▼

start

Stats:

Process failed!

Process result.

Output path:

log

⋮